

Incorporating UCD

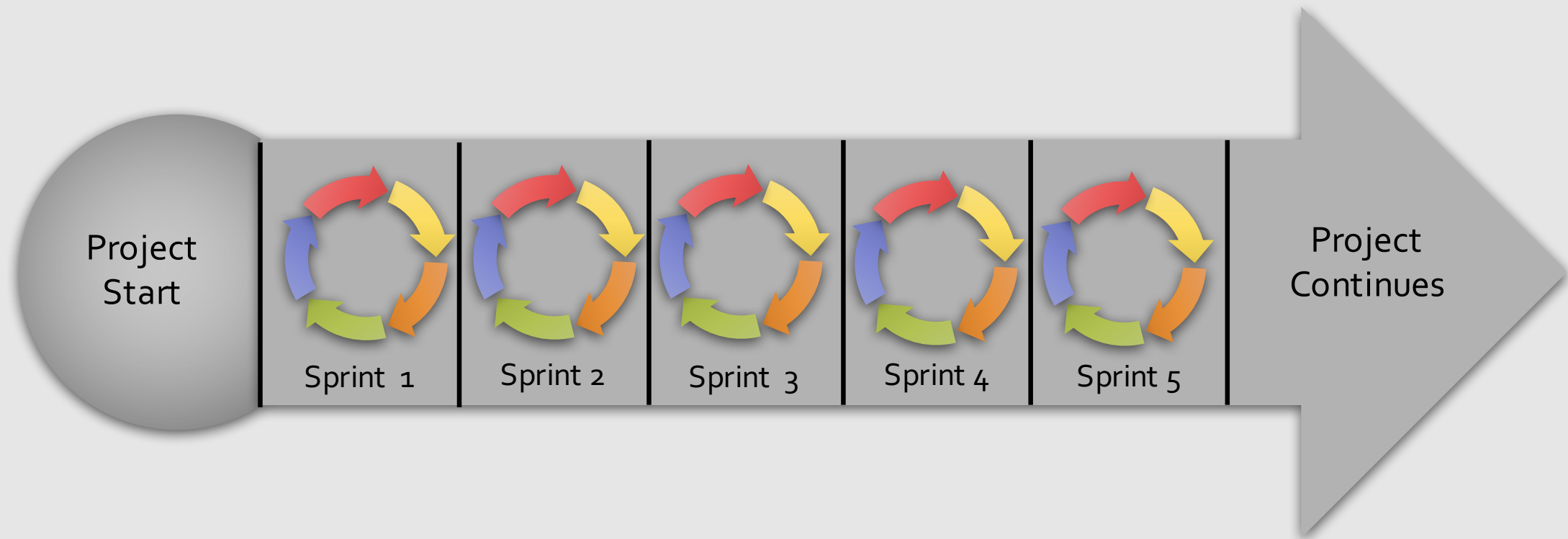
Experience is the product

UCD Philosophy
Creatives

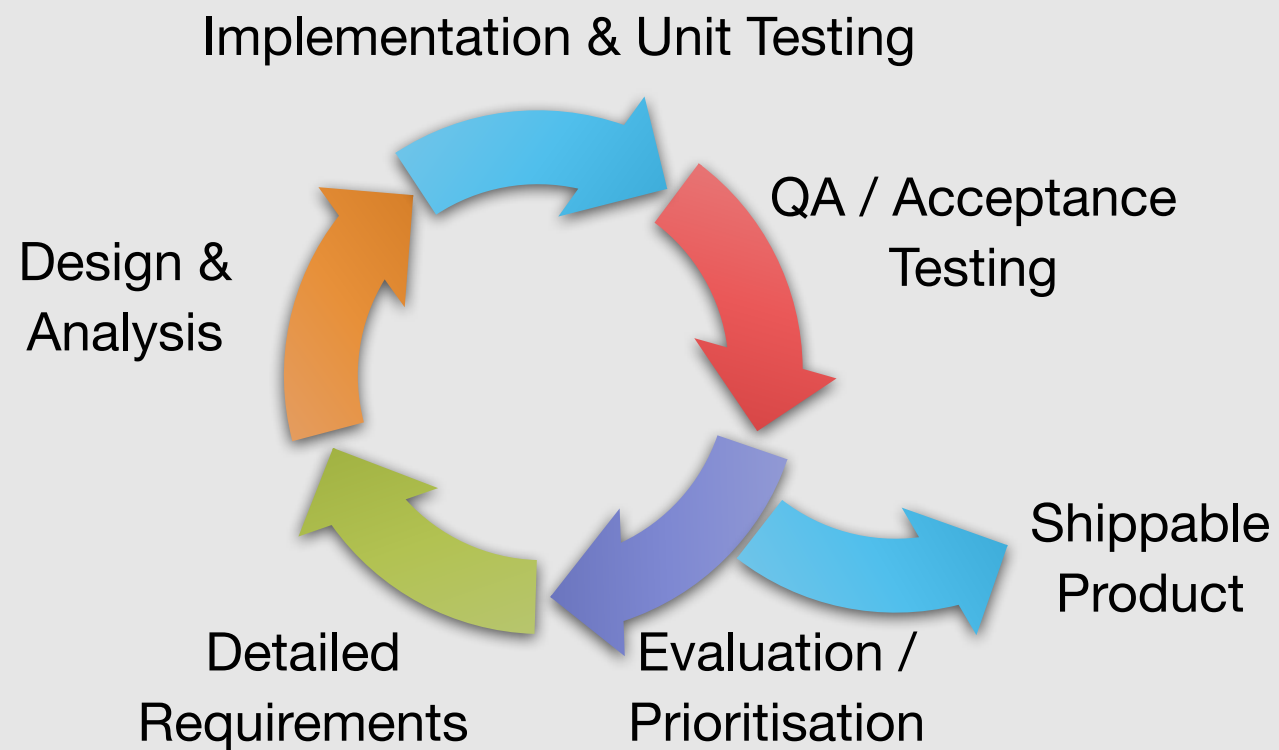
“Experience is the product”

Agile Philosophy
Developer Led
Deliver working software

Scrum Methodology



Iteration Detail



User Centered Design

Get out of the building

Learning

- User Centered Design
 - UCD Methods

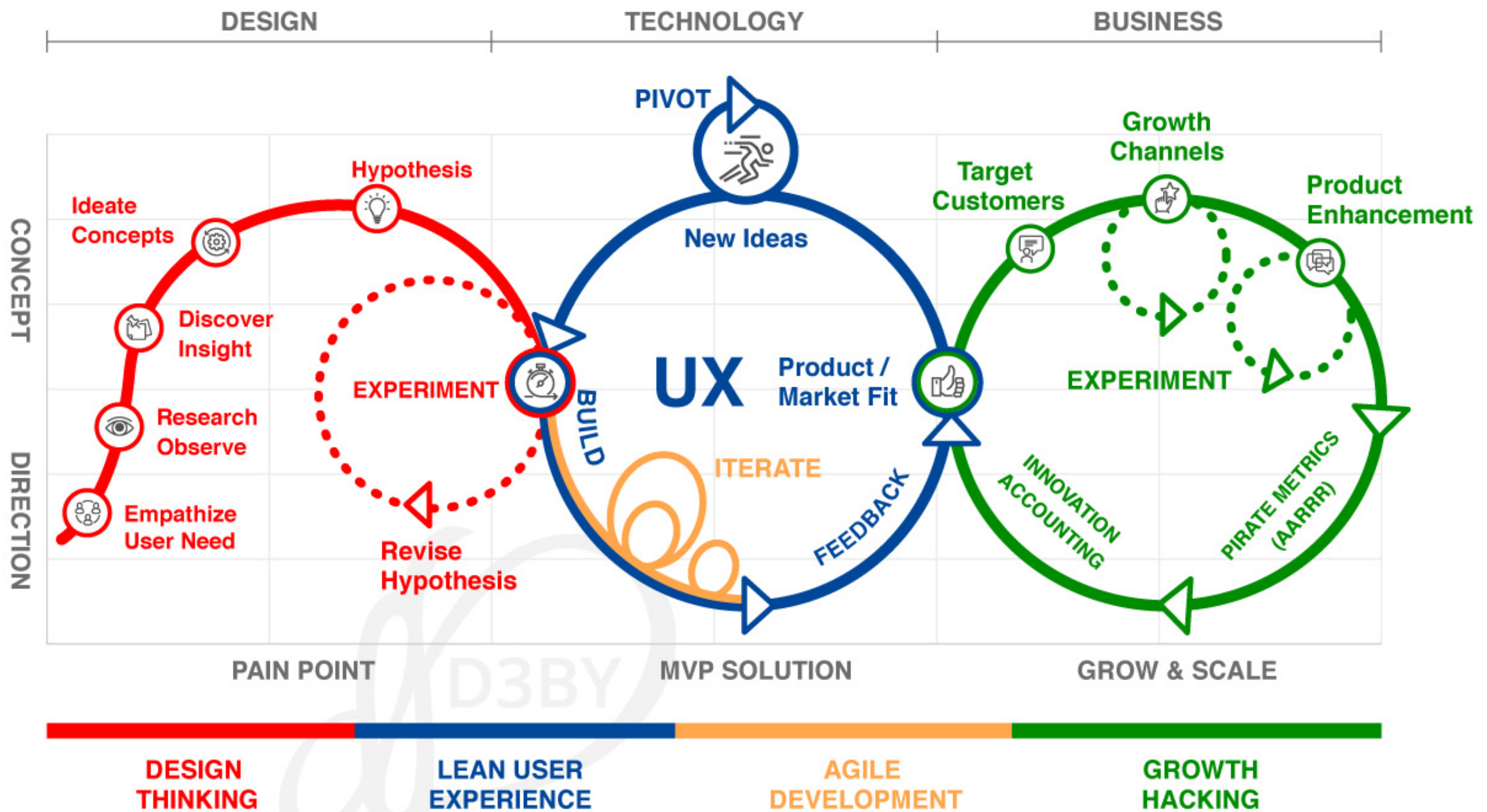
Motivation

- Improve the product
- Continue to compete
- “Users will evaluate your interface sooner or later” (Hix & Hartson, 1993)

User Centred Design

User-centred design is a process in which the needs, wants, and limitations of end users of a product are given extensive attention at each stage of a products design process.

- Requirements gathering
 - Understanding and specifying the context of use
- Design
 - Producing designs and prototypes
- **Testing & Evaluation**
 - Perform user based testing & evaluation
 - Usability testing



UCD / Agile similarities

- Cyclic
- Progressing in steps
- Validation and refinement during the process

UCD versus Agile (Conflicts)

- Agile is value driven

“Our highest priority is to satisfy the customer through early and continuous delivery of working software”

- UCD is UX focused

“Our highest priority is to help create an experience for end-users where they can achieve their goals easily and efficiently with minimal disruption to their mental model of the problem space”

UCD versus Agile (Conflicts)

- Agile

“Deliver working software frequently, from a couple of weeks to a couple of months, with a preference to the shorter timescale”

- UCD

“With UCD, software that simply works is a secondary measure of progress. Much more important is whether the software helps users achieve their goals”

UCD and Agile Integration Issues

- Divided teams
 - You spec and design it and we will build it
 - Work toward the shared vision of success
- Handover is the enemy of understanding

“The most efficient and effective method of conveying information to and within a development team is face-to-face conversation”
- No time to iterate
 - Agile teams don't plan to iterate
 - UCD promotes iterations

SCRUX

Day 1:

- Sprint planning AM - 1-2 hours
- Sketching - 6 hours

Day 2

- Stand up: sketch review AM - 30 minutes
- Prototyping wireframes - 7 hours
- Visual design exploration - 7 hours

Day 3

- Stand up: AM - 15 minutes
- Prototyping wireframes - 7 hours
- Visual design templates - 7 hours

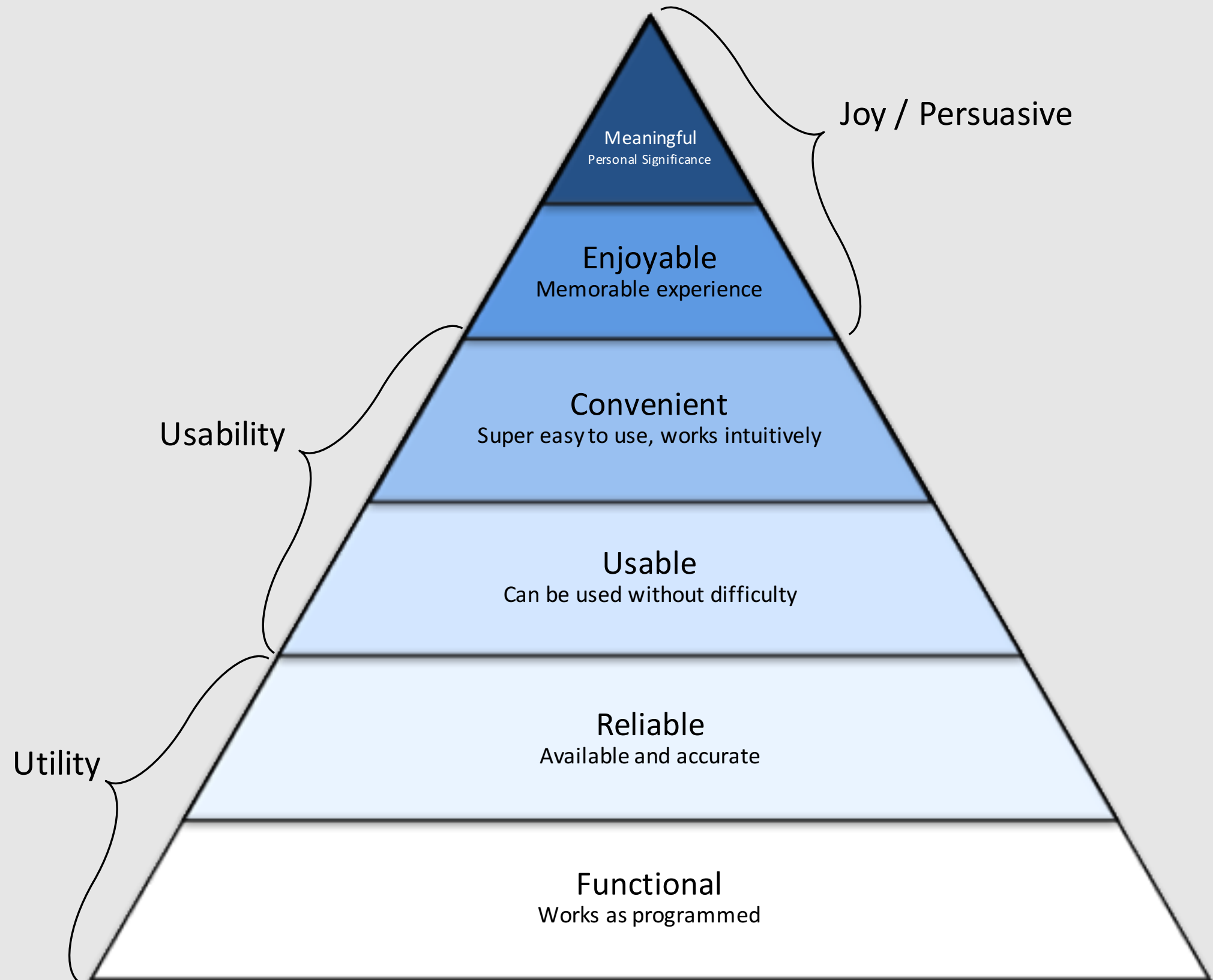
Day 4

- Stand up: AM - 15 minutes
- Prototyping wireframes - 7 hours
- Visual design templates - 7 hours

Day 5

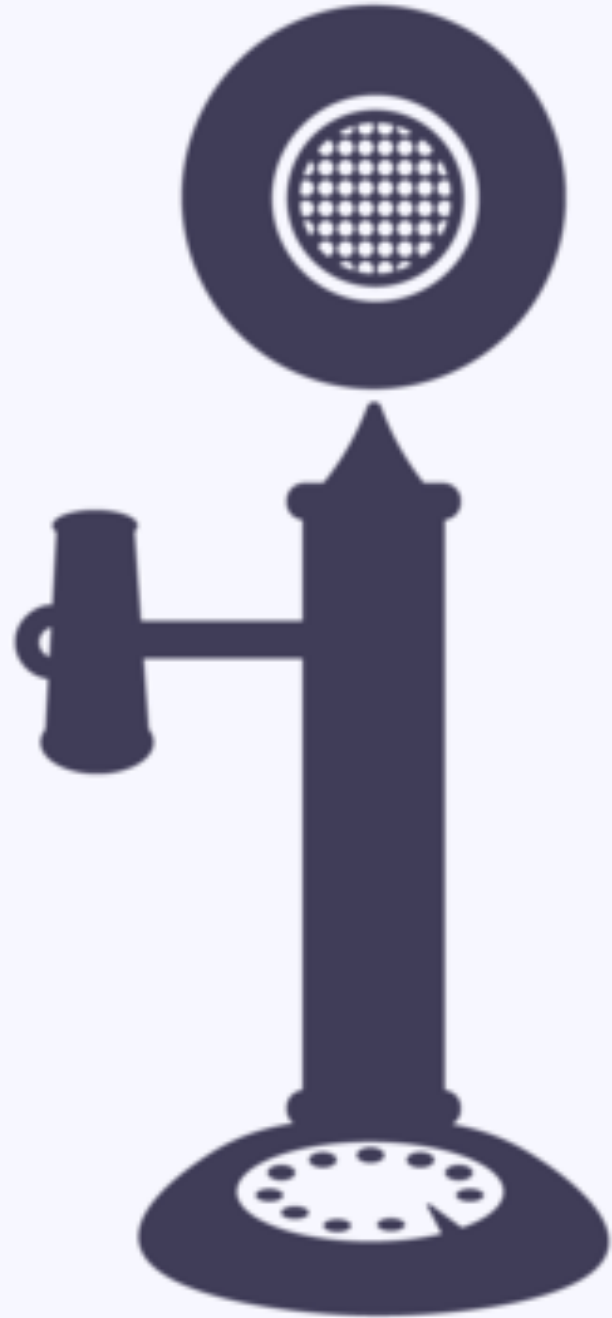
- Sprint Review: AM - 2 hours
- Prototype amends - 7 hours
- Visual design templates - 7 hours
- Retro - 30 minutes

Product Evolution

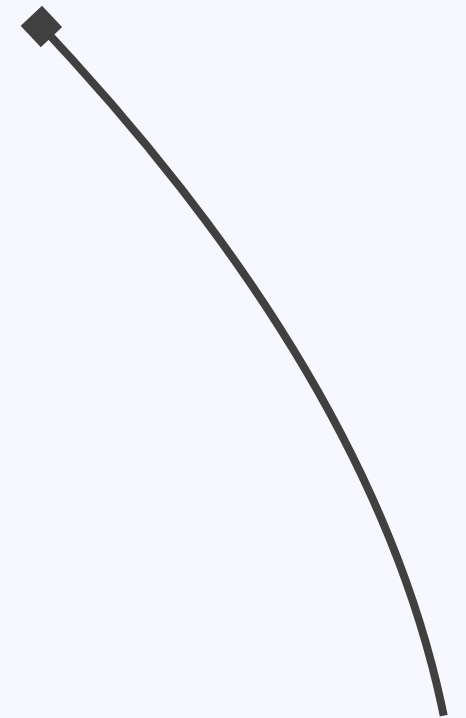


Persuasive

Usability



Utility



Usability - Definition

ISO/IEC FDIS 9126-1

Usability

The capability of the software product to be understood, learned, used and attractive to the user, when used under specified conditions.

- Easily understood, intuitive
- Learned – not complicated
- Usable under context – Environment is important!

Usability - Definition

ISO 9241-11: Guidance on Usability (1998)

Usability

The extent to which a product can be used by specified users to achieve specified goals with **effectiveness**, **efficiency** and **satisfaction** in a specified context of use.

- Effective
- Efficient
- Satisfactory

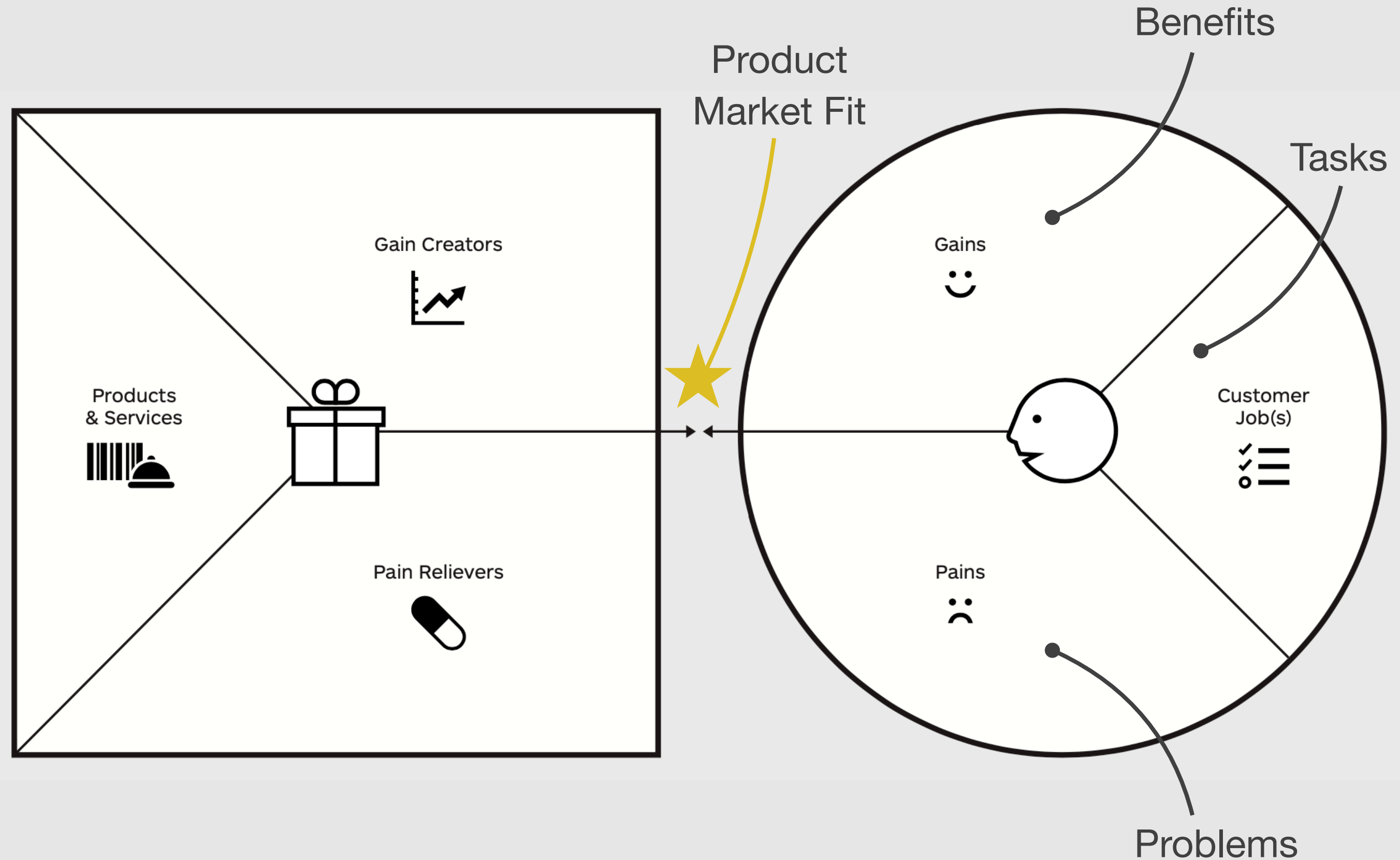
User Centred Design

You are not your user

UCD Methods

- **Value Proposition - business centric**
- Red Routes
- Field Studies
- Focus Groups
- Card sorting
- Participatory design
- Personas
- Questionnaires
- Interviews
- Usability testing
- Heuristic Evaluation and Expert Review
- Eye Tracking

Value Proposition Canvas



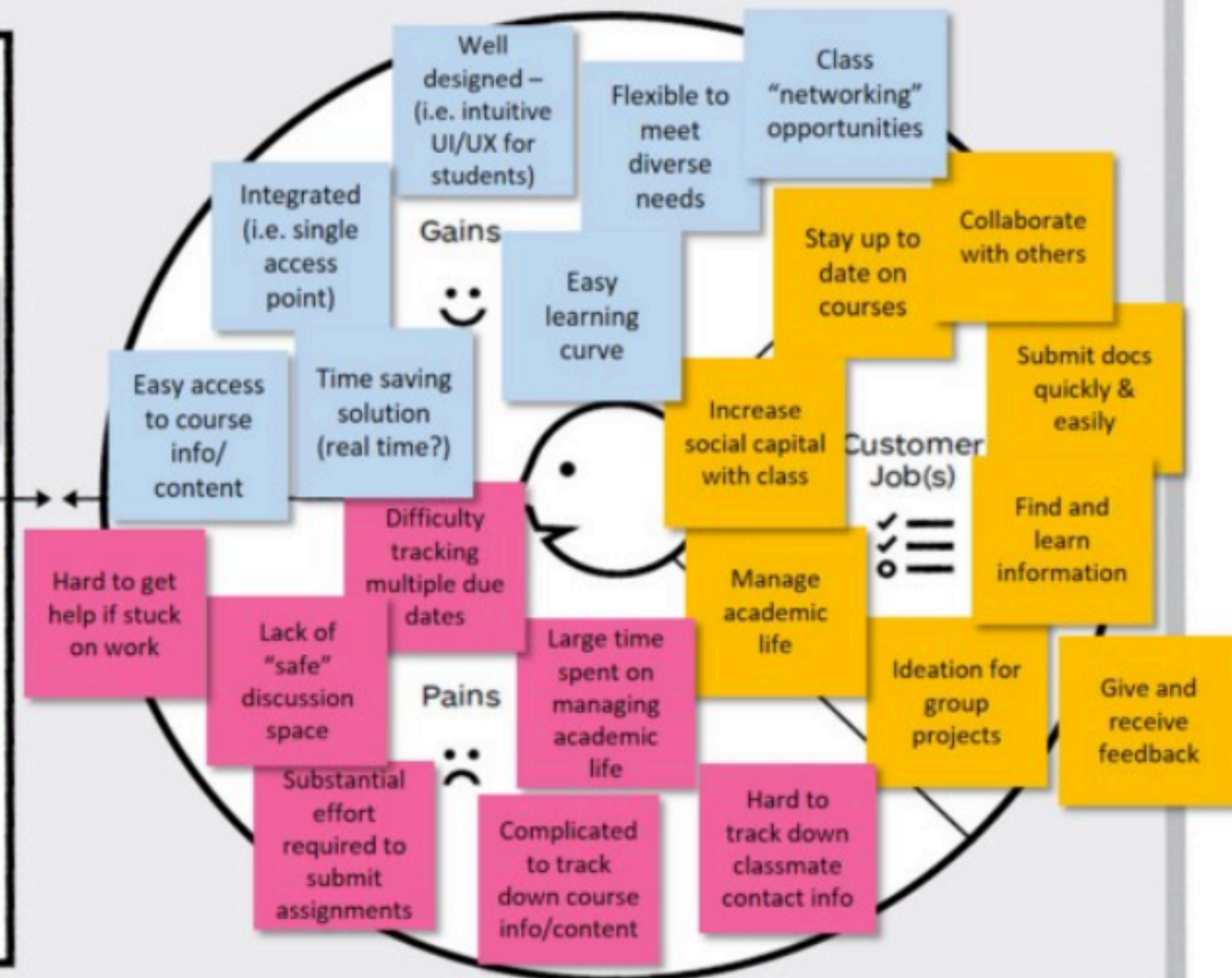
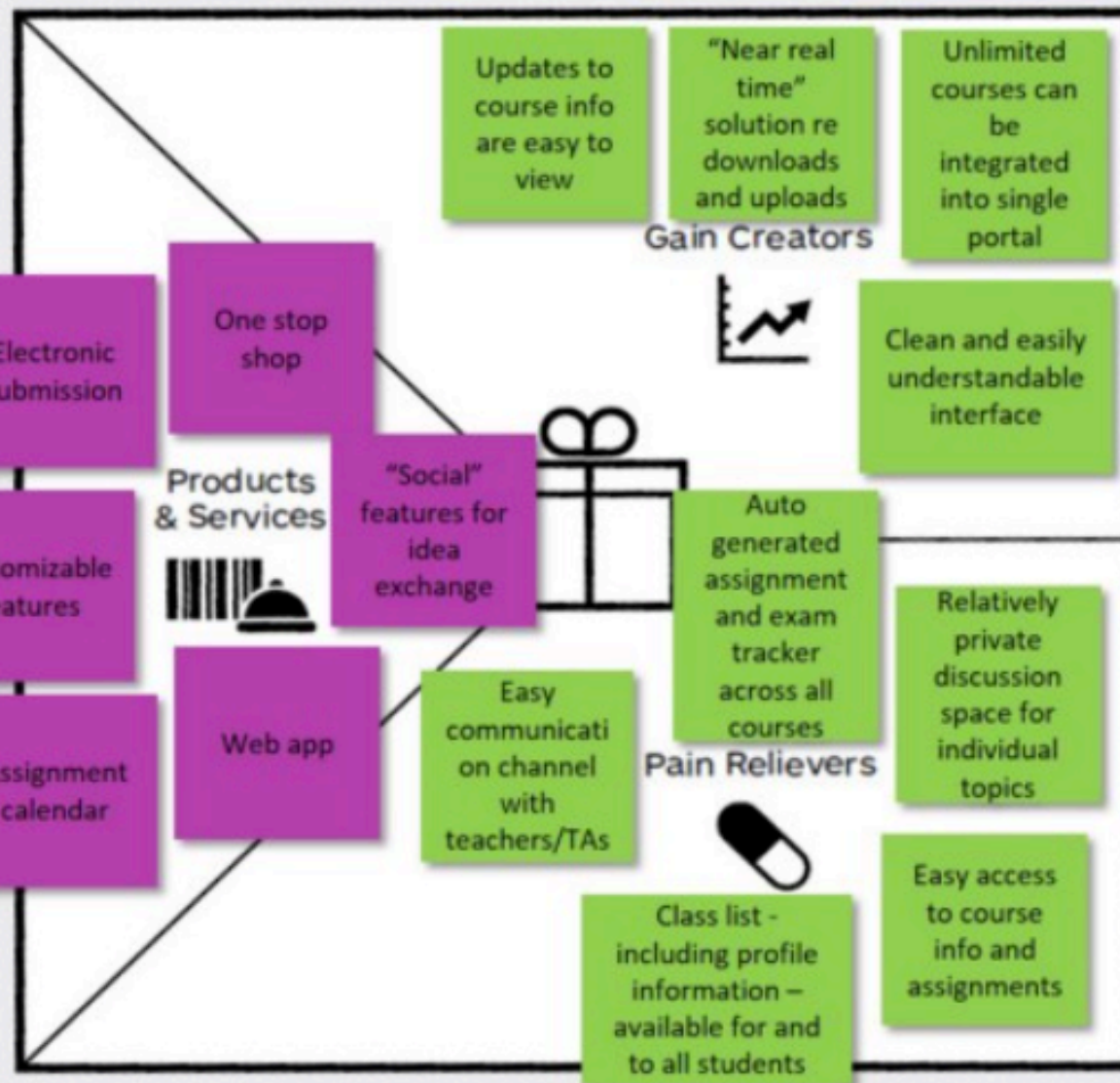
VPC Steps

1. Choose a customer segment i.e music tutor
2. Identify their jobs and prioritise them according to how important they are to your customer.
3. Identify pains and prioritise them.
4. Identify their gains and prioritise them.
5. Pick the top 3-5 most important pains and gains that relate to the most important jobs.
6. Create a list of features of your product or service.
7. List all pain relievers.
8. List of gain creators.
9. Pick 3-5 of the gain creators and pain relievers, that make the biggest difference to your customer.
10. Link the pain relievers, gain creators and product benefits to the pains, gains and jobs they solve.

The Value Proposition Canvas

Value Proposition Single platform to share content, and interact with peers + teachers, for all registered courses

Customer Segment Harvard students



UCD - Red Routes

- A red route is a **frequent** or/and **critical activity** that users perform!
 - User perspective
 - Business perspective
 - Examples
 - Perform a search - frequent
 - Update your bank card details - critical
 - Check out shopping cart - frequent/critical
- Needs to be unobstructed
- Quickly achievable with minimal effort
- Identify and optimise red routes

Red Routes

UPDATE

Here's another bad password policy, courtesy of TechRepublic:

Your password may be any combination of 6 to 10 letters and numbers.

- It is case sensitive
- It can't contain special characters (?&%\$#@+=!~ , etc.)
- It can't contain your user name
- It can't contain two separated numbers (i.e., Abc12ef34 would be invalid)
- It can't contain "Netnet" or "Password"

Can't contain two separated numbers? I don't even. If you've run across other examples like these, tweet at me.

Card Sorting

- Helps with input to design
- Users are presented with a pack of cards
- Each card contains a statement
- Users are asked to categorise these
 - Free categories (user decides)
 - Place into predetermined categories
- Helps assign content into categories

Card Sorting - Example



Field Studies

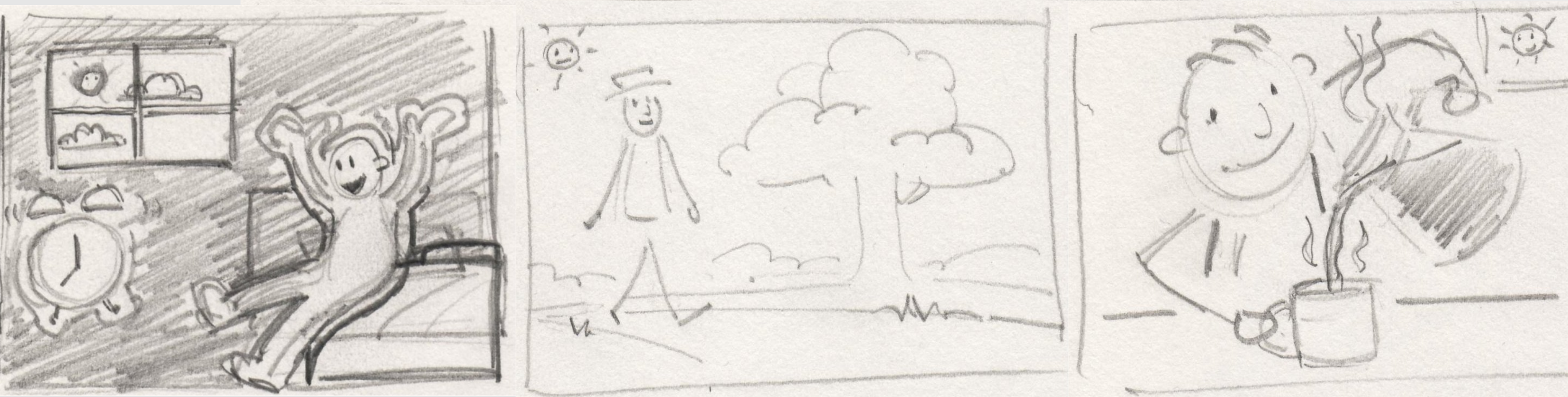
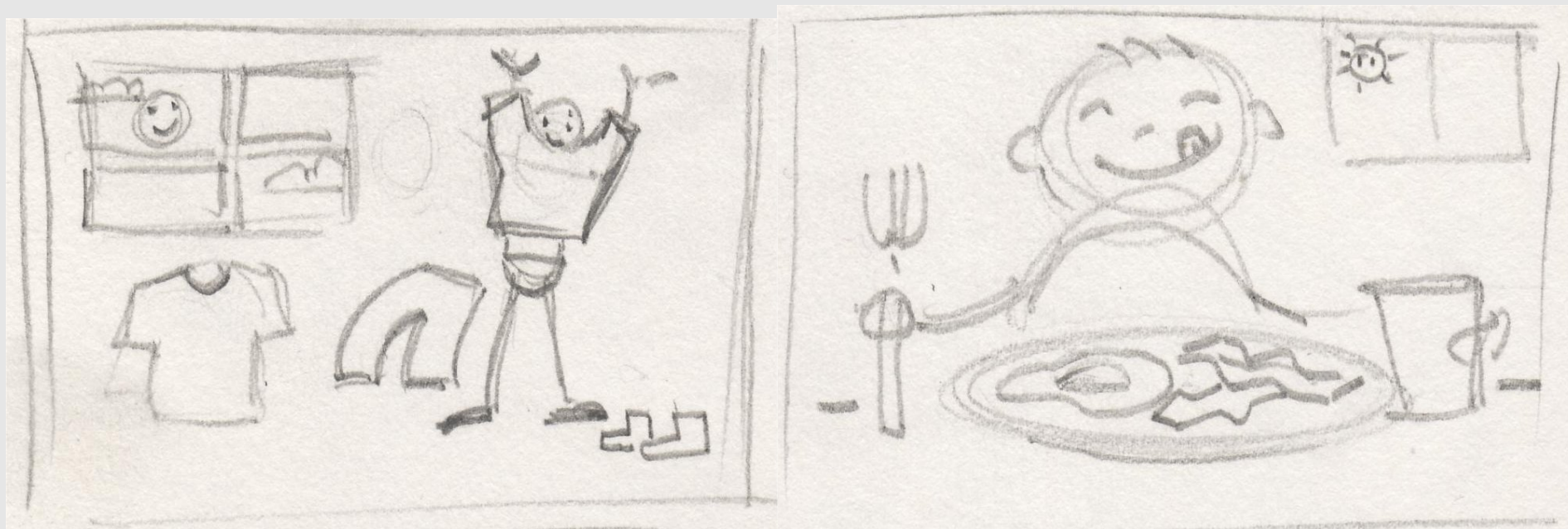
- Immerses the designer into the users environment
- Gains unique insight/information
- Shadow the user
 - Akin to studying animals in the wild
- Attention to details
- Easy to observe
- Expensive
- Learn
 - User & their environment

Focus Groups

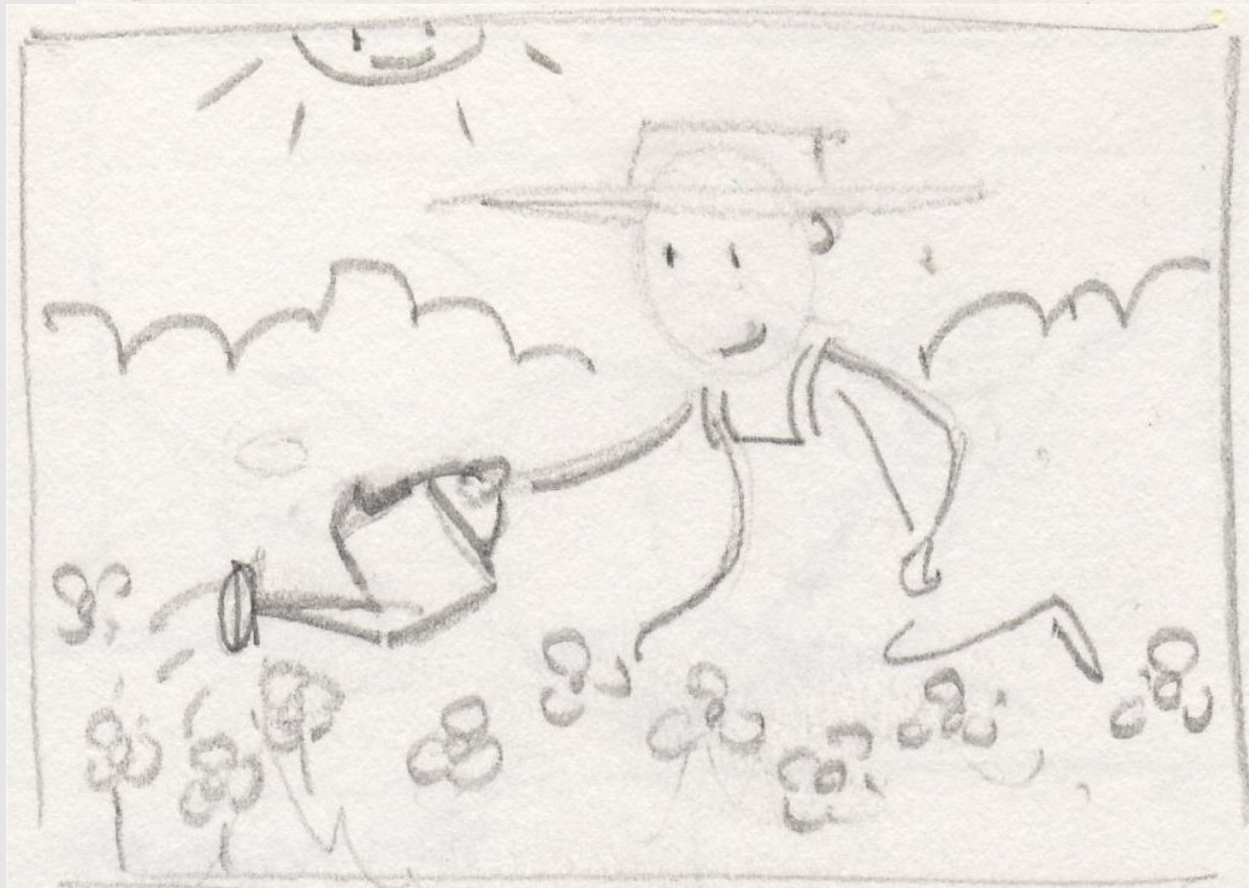
- Small group of participants (8-12)
- Sharing ideas, thoughts, feelings and attitudes
- Get valuable input prior to the design
- Led by an Expert



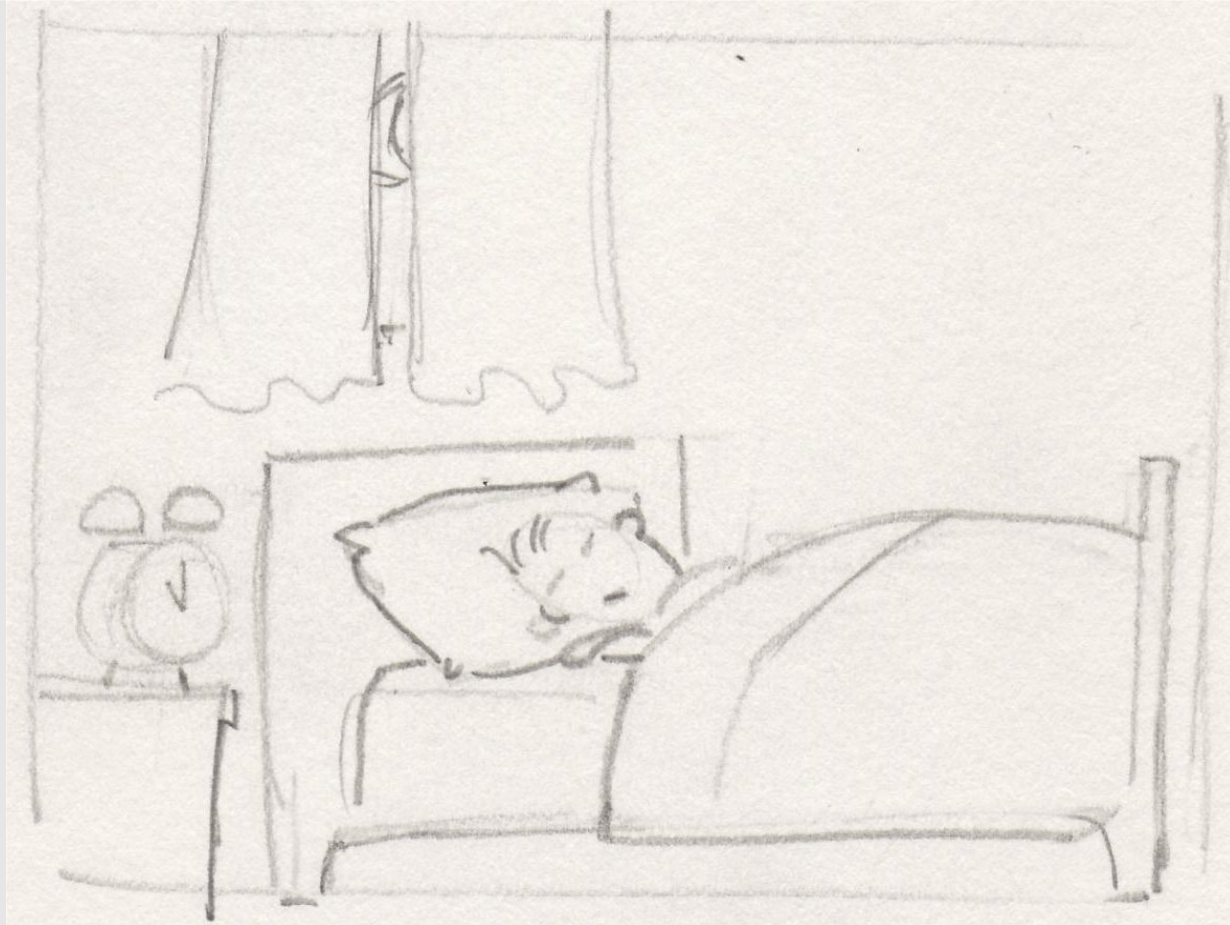
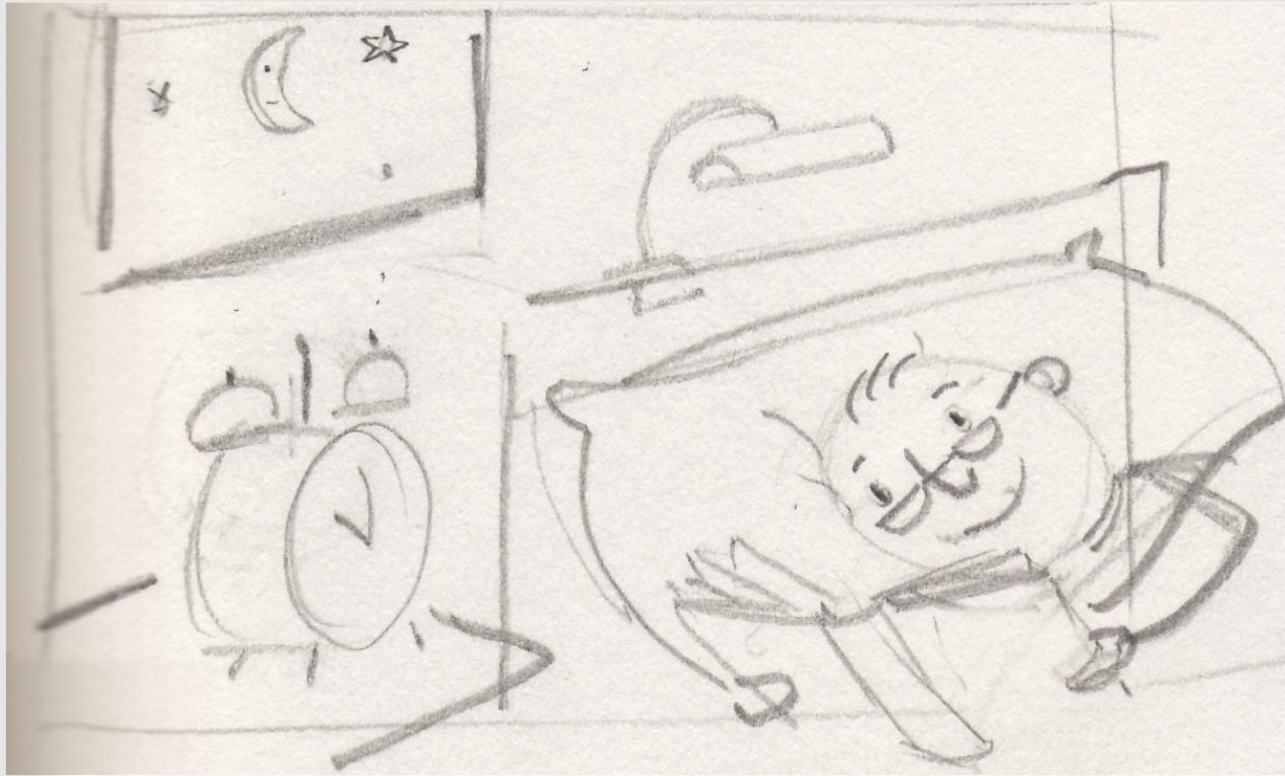
Example Focus Group



Example Focus Group



Example Focus Group



Questionnaires

- Only used when remote access to users is available
- Useful to gain insight from a larger sample size
- Easy to administer
 - Remote access
 - Cheap
 - Easy to Implement
- Allows the design team to raise questions
- Can be difficult to design
 - Must be clear, well designed
 - Non-bias
- Quantitative & qualitative



Questionnaires - Biases

- Avoid leading questions
 - What do you think of the football legend George Best
- Provide mutually exclusive choices
- Cover all exhaustive answers or provide “Other” option
 - What age are you? 0-10, 11-18, 22-30, 30+
- Use balanced scales
 - What do you think of Medicare?
 - Pretty good, Great, Fantastic, Incredible, Best ever
- Ask only one question at a time
 - What is the fastest and cheapest way to travel to uni?

Interviews

- One to one - unique point of view
- Output from the interview must be carefully analysed or recorded
- Used to gather requirements i.e early stage
- Could perform evaluation
- Useful to dig into the why and any negative issues
- Expensive
 - Travel
 - Time

Participatory Design

- Users are actively involved in the design process
- Usually produces a prototype
- Prototype feeds into overall design process
- Designers, developers and users get together in a workshop
- Requires an expert to guide the process

Overview of Methods

Method	Cost	Output	Sample size	When to use
Field Test	High	Non-statistical	Low	Requirements gathering & evaluation
Focus groups	Low	Non-statistical	Low	Requirements gathering
Usability testing	High	Statistical & Non-statistical	Low	Design & Evaluation
Card Sorting	High	Statistical	High	Design
Participatory design	Low	Non-statistical	Low	Design
Questionnaires	Low	Statistical	High	Requirements gathering & evaluation
Interviews	High	Non-statistical	Low	Requirements gathering

Heuristic Evaluation

- Usability experts review your product
- Compared against accepted usability standards
- Produces list of usability issues
 - Different from that produced using Usability testing

Heuristic Evaluations

Advantages

- Quick and relatively inexpensive feedback
- Feedback obtained early
- Used to complement other usability testing methodologies

Disadvantages

- Requires knowledge and expertise
- Usability experts are hard to find and can be expensive
- Need to use multiple experts to aggregate results
- Identifies more minor issues and few major issues

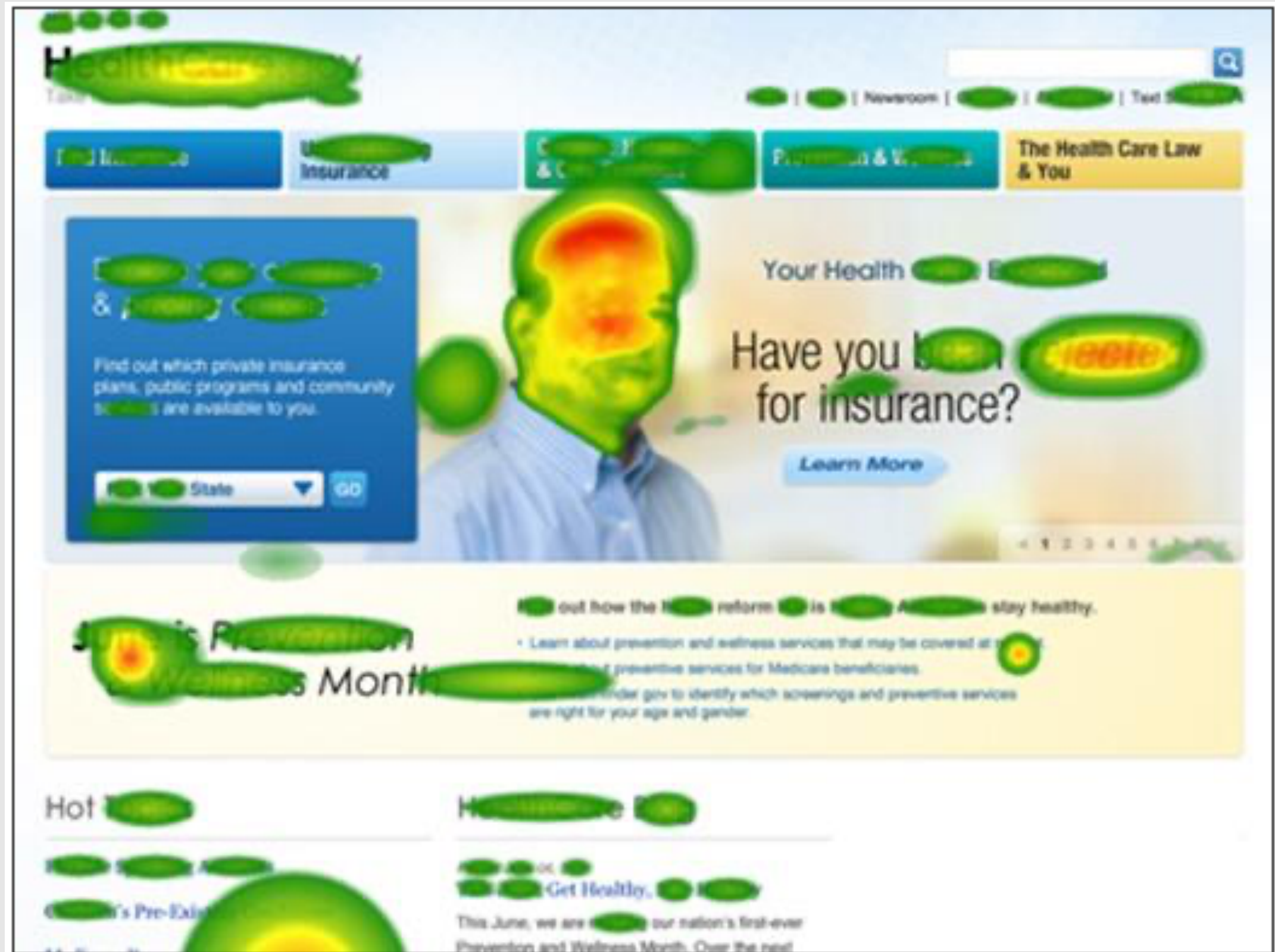
Heuristic Evaluation

- Neilson Group Heuristics
 - Visibility of system status
 - Match between system and real world
 - User control and freedom
 - Consistency and standards
 - Error prevention
 - Recognition rather than recall
 - Flexibility and efficiency of use
 - Aesthetic and minimalist design
 - Help user recognise, diagnose and recover from errors
 - Help and documentation

Eye Tracking

- Purpose
 - Where user look
 - How long they are looking
 - Moving from item to item
 - What's missed

Eye Tracking - Heat Maps



Eye Tracking

- Are users reading or scanning
- Shows users intensity of attention
- Pupil dilation determines if they are searching for something i.e a keyword
- Good for comparing user groups
- Cannot tell you why
- Not 100% effective in all cases
 - Glasses, contact lenses etc
- Can't be certain about seeing things or not.

First Click Testing

First Click Testing examines what a user would click on first on the interface in order to complete their intended task. It can be performed on a functioning website, a prototype or a wireframe.

- A participant who clicks down the right path on the first click will complete their task successfully 87% of the time.
- A participant who clicks down the wrong-path on the first click, tends to only successfully complete their task 46% of the time.

Usability Testing

- Asks users to complete a series of tasks - structured
- Start (wireframes), Middle (prototypes) or End (product)
- Monitor the process
 - Use the think aloud process
 - Eye tracker
- Monitored by two individuals
 - Moderator
 - Note taker/observer

Further Reading

- <https://www.nngroup.com>
- <https://www.usability.gov/>
- <https://usabilitygeek.com>
- www.humanfactors.co.uk
- www.uxmag.com
- <https://www.thoughtworks.com/insights/blog/getting-agile-user-centered-design>
- SCRUX - <https://www.webcredible.com/blog/how-apply-strict-agile-process-ux-design-introducing-scrux/>
- [Thinking Hats](#)